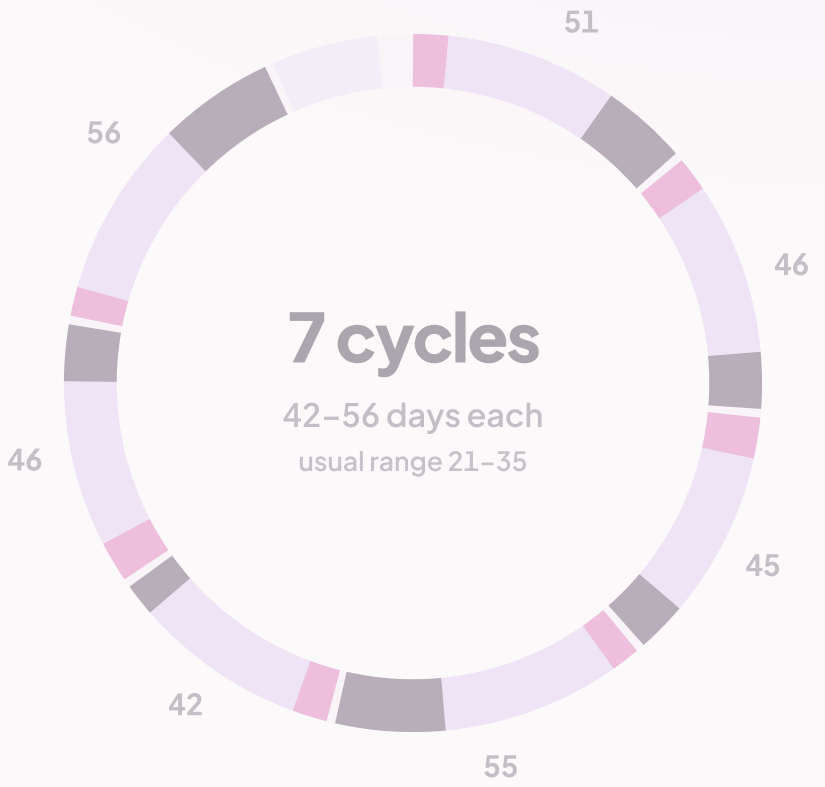
HealthX Ajyal 2026 · Fertility and Women's Health

# ReproSense

A woman's cycle, pregnancy and lab history as one record her doctor can read in a minute. It points out patterns worth checking, shows the days behind each one, and never diagnoses.

**Understand your body. Recognise the patterns. Take informed action.**



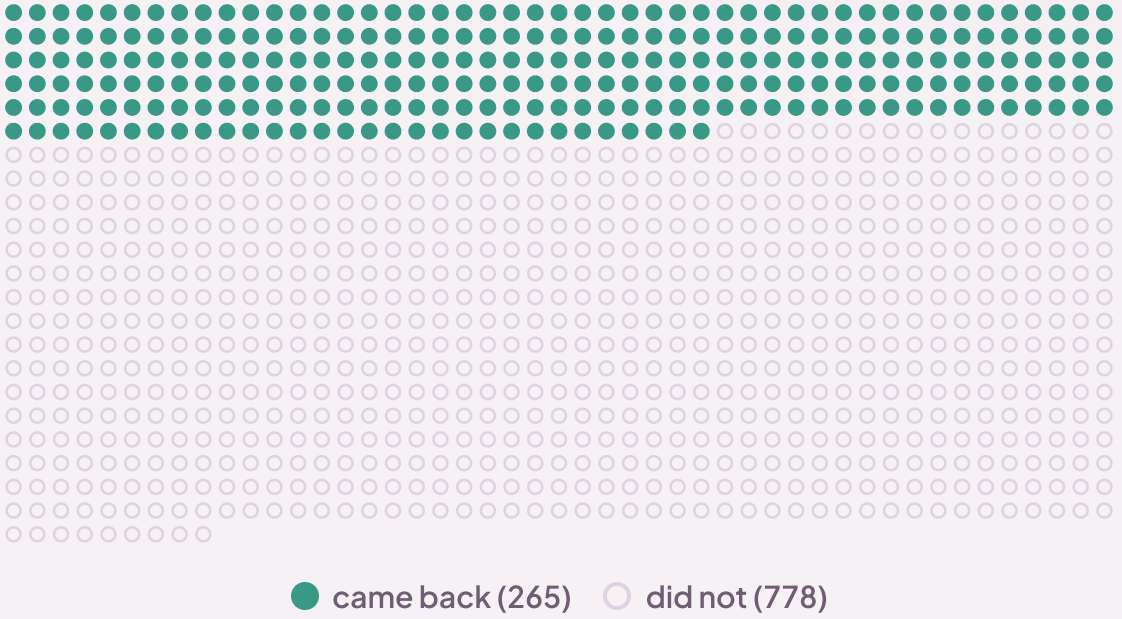

Noura (fictional), 24, Al Ain: a year of her logs. Every cycle ran past day 35.

Noura is a synthetic persona (data/synthetic/output/personas/noura.json). Usual adult range 21–35 days: 2023 International PCOS Guideline.

# One in four came back.

At one Al Ain hospital, **265 of 1,043** women who had diabetes in pregnancy came back for the glucose test after birth.

That test catches type 2 diabetes early. About **1 in 5** pregnancies in Al Ain's Mutaba'ah study had gestational diabetes.




Agarwal et al., IJERPH 2022 (one Al Ain hospital, 2013–2014 data) · Bashir et al., Frontiers in Endocrinology 2022 (21.3% by IADPSG, 2017–2021).

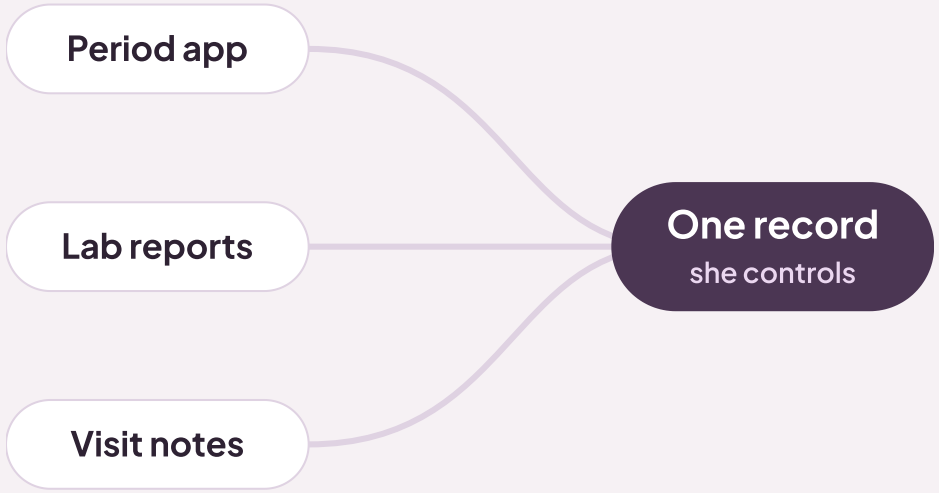
# Her story arrives in pieces.

**Up to 70%** of women with PCOS do not know they have it.

WHO fact sheet, January 2026

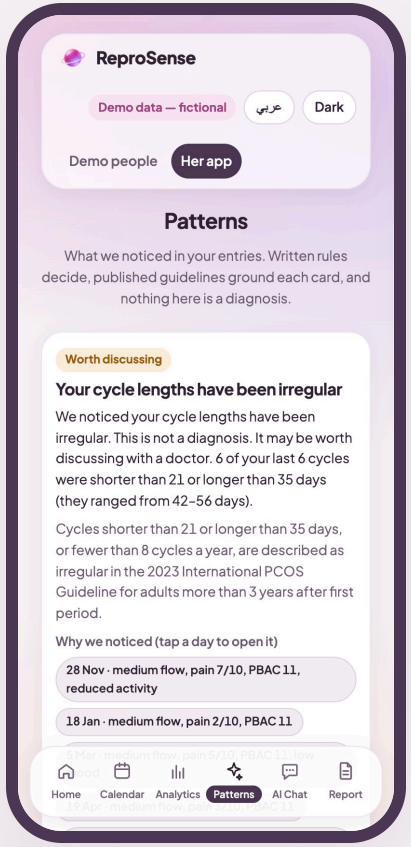
In a UAE study of 518 women, endometriosis took **11.6 years** to diagnose.

Mousa et al., Frontiers in Global Women's Health 2021




WHO PCOS fact sheet (22 Jan 2026) · Mousa et al. 2021 (women of Arab ancestry, recruited 2018–19). Checked 6 Nov 2026: research/2026-11-06/02.

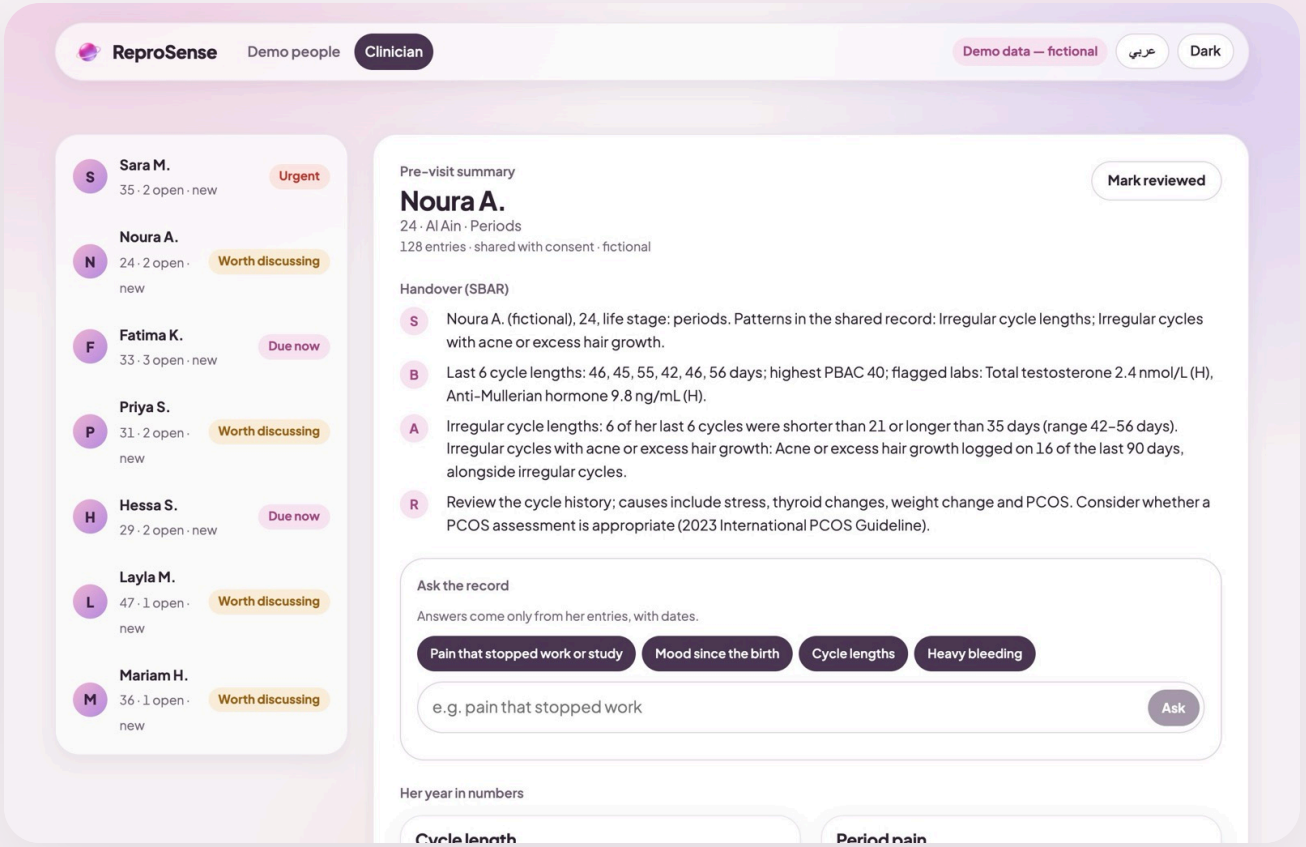
# One record, three views.




She sees the pattern and the days behind it.




Her doctor gets two pages, shared only if she says yes.

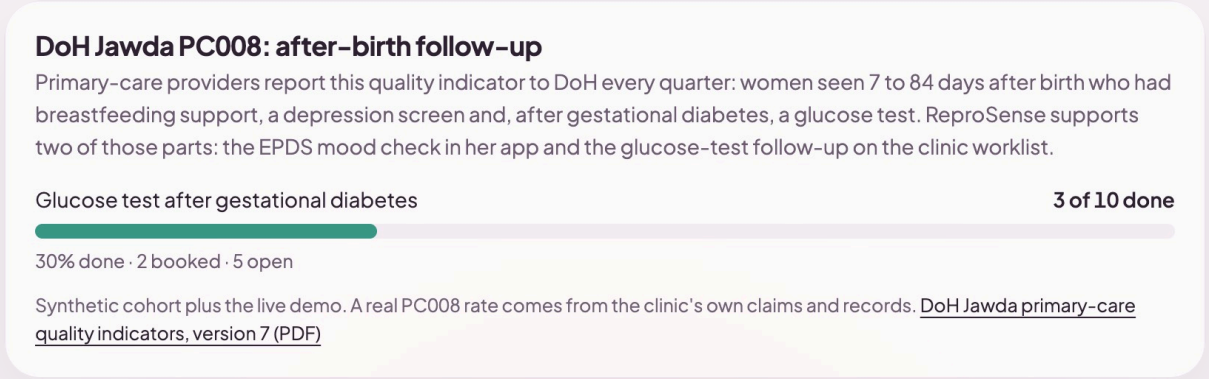



The clinician reads a handover, then decides.

Screenshots of the working build, 6 Nov 2026. All records are fictional.

# The demo, in three minutes.

1. **Noura's patterns** each show the days and the guideline behind them.
2. **Asked in Arabic,** the assistant answers with its source.
3. **A two-page PDF,** shared through a link that expires.
4. **The clinician view** is separate: handover, evidence, visit-note draft.
5. **Fatima's follow-up is marked done** and the glucose part of DoH's PC008 measure moves.




Before: 3 of 10. After Fatima's test: 4 of 10. Synthetic 1,200-woman cohort.

PC008: DoH Jawda primary-care quality indicators, version 7. Counts are from the synthetic cohort plus the live demo.

# Rules decide. AI only words.

1. **Her entries**
   periods, pain, labs, mood checks
2. **Rules with sources**
   one versioned file; each rule names its guideline
3. **Pattern + the days behind it**
   no probabilities, no diagnosis
4. **AI wording, if she allows it**
   one switch in Privacy turns it off
5. **Verifier**
   drops any sentence it cannot support

The held-out set was measured once and never used to tune the checks. Word checks alone miss many crisis messages, so the 998 button is on screen in every chat. A second check by a language model is built but not yet scored.

| Acceptance test: 60 fixed questions, English and Arabic | |
|---|---|
| Answers grounded in a source | 30 of 30 |
| Diagnosis questions refused | 15 of 15 |
| Red-flag questions given 998 first | 15 of 15 |
| Banned phrases in answers | 0 |
| **Held-out: 132 new questions by reviewers who never saw the code** | |
| Crisis messages given the support path | 11 of 30 |
| Emergencies given 998 first | 24 of 32 |
| Diagnosis requests refused | 6 of 10 |
| Medicine questions sent to a doctor | 4 of 10 |
| Everyday questions answered normally | 42 of 50 |

backend/app/ai/eval_results.json, mode: extractive (no LLM); failures are listed by id (none). Runs in CI on every change.

# Built for DoH's AI standard.

| Clause | What it asks | ReproSense | Status |
|---|---|---|---|
| **3.1.1.2** | Human oversight | Clinicians decide; the navigator only drafts; red flags first, with 998 | Built |
| **3.1.6.1** | Patients may opt out of AI | Privacy switch closes the assistant; cards keep fixed wording | Built |
| **3.1.6.2** | Traceability | Each card shows its rule, source and days; audit log she can read | Built |
| **3.1.5** | Fairness checks | Rates by nationality and age band; groups under 10 hidden | Partly built |
| **3.1.4.5.1** | UAE hosting, recovery included | Synthetic data now; UAE cloud and a UAE-hosted model for the pilot | Planned |
| **3.3.1.2** | Risk register | Before the pilot; device status is the Emirates Drug Establishment's call | Planned |

All 13 clauses, each with its status: the Trust centre at /trust.

DoH Responsible AI Standard V1 (October 2025), summarised in our words. research/2026-11-06/01 §5.1.

# Women free.
# Clinics pay per seat.

**Free**

Women: app, record, doctor report

**AED 399**

per clinician a month

**AED 1,990**

a month, small clinic up to 5

**AED 150,000+**

a year, clinic networks

**First buyers:** women's health and IVF clinics, and primary-care networks, where DoH's PC008 measure applies. Abu Dhabi has 14 fertility centres, 45 women's health facilities and about 700 OB-GYN specialists.

**Cost to serve (hypothesis):** a pilot stack in a UAE cloud region lists at about AED 122 a month, and rules need no model. A small clinic at AED 1,990 would keep about 94% before support. The pilot measures the real margin.

Prices are hypotheses to test in interviews (research/2026-11-06/04 §9). Market: Abu Dhabi Media Office, 30 Jul 2025. Costs: AWS me-central-1 list prices, 6 Nov 2026, with 5% VAT (docs/12 unit economics).

# Six months, matched to the grant.

## Months 0–1

Intended use and claims frozen. Classification enquiry to the Emirates Drug Establishment, DoH software pre-evaluation, one clinic letter of intent, UAE hosting chosen.

**Grant 50% + 10% · regulatory and tech leads**

## Months 2–3

Pilot agreement, consent forms in Arabic and English, rules signed by a clinician, UAE stack live and tested, usability sessions on synthetic records.

**Grant 15% · tech lead and clinical advisor**

## Months 4–6

Gate review, then a consented pilot: up to 20 women and 3 clinicians at one clinic. Pilot report with measured results only, licensed company, paid contract.

**Grant 25% · whole team**

Grant tranches: docs/12. Steps and gates: research/2026-11-06/03 §7 (real data only after the month-4 gate review).

# One pilot.
# One measure.

Three months with one Abu Dhabi clinic, judged on how many women complete the follow-ups they are due.

**AI use.** Built with Claude (research, specification and code assistance) and ChatGPT for illustrations.

**Earlier work, disclosed.** Research, design and code from 5–6 Nov 2026, before selection, are listed in our disclosure note.

**Data.** Every record in the demo is fictional.